

Problem

A balanced wye-connected load with a phase impedance of $10 - j16 \Omega$ is connected to a balanced three-phase generator with a line voltage of 220 V. Determine the line current and the complex power absorbed by the load.

Step-by-step solution

Step 1 of 2

The wye impedance is

$$\mathbf{Z}_Y = 10 - j16 \Omega.$$

The line voltage is

$$\mathbf{V}_{ab} = 220 \text{ V}.$$

$$\mathbf{V}_{ab} = \sqrt{3} V_p \angle 30^\circ.$$

Determine the line current.

$$\begin{aligned} \mathbf{I}_a &= \frac{\mathbf{V}_{an}}{\mathbf{Z}_Y} \\ &= \frac{V_p \angle 0^\circ}{\mathbf{Z}_Y} \\ &= \frac{\mathbf{V}_{ab} \angle 0^\circ}{\sqrt{3} \angle 30^\circ} \\ &= \frac{\mathbf{V}_{ab}}{\sqrt{3} \angle 30^\circ} \end{aligned}$$

Substitute 220 V for $\mathbf{V}_{ab}$.

$$\begin{aligned} &= \frac{220 \angle 0^\circ}{\sqrt{3} \angle 30^\circ} \\ &= \frac{127.02 \angle -30^\circ}{18.87 \angle -58^\circ} \\ \mathbf{I}_a &= 6.73 \angle 28^\circ \end{aligned}$$

Therefore, the line voltage is $\boxed{\mathbf{I}_a = I_p = 6.73 \text{ A}}.$

Step 2 of 2

Determine the complex power absorbed by the load.

$$\begin{aligned} \mathbf{S} &= \sqrt{3} V_L I_L \angle \theta \\ &= \sqrt{3} (220) (6.73) \angle -\tan^{-1} \left(\frac{16}{10} \right) \\ &= 2564.5 \angle -58^\circ \\ \mathbf{S} &= 1358.98 - j2174.82 \end{aligned}$$

Therefore, the complex power absorbed by the load is $\boxed{\mathbf{S} = 1359 - j2175 \text{ VA}}.$